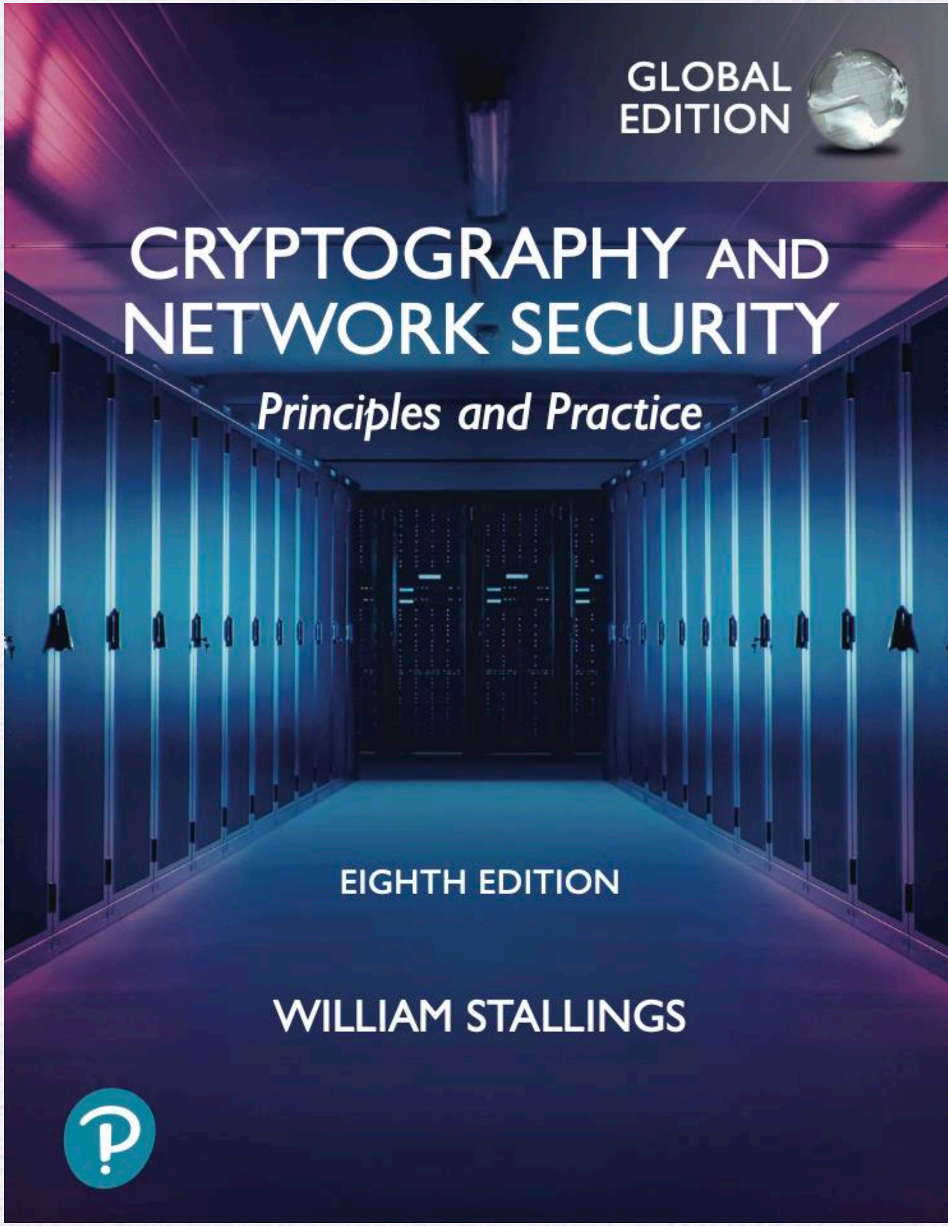
University of Nevada – Reno
Computer Science & Engineering
Department

CS454/654 Reliability and Security
of Computing Systems - Fall 2025



CHAPTER 16

User Authentication

16.1 Remote User-Authentication Principles

The NIST Model for Electronic User Authentication
Means of Authentication
Multifactor Authentication
Mutual Authentication

16.2 Remote User-Authentication Using Symmetric Encryption

Mutual Authentication

16.3 Kerberos

Motivation
Kerberos Version 4
Kerberos Version 5

16.4 Remote User-Authentication Using Asymmetric Encryption

Mutual Authentication
One-Way Authentication

16.5 Federated Identity Management

Identity Management
Identity Federation

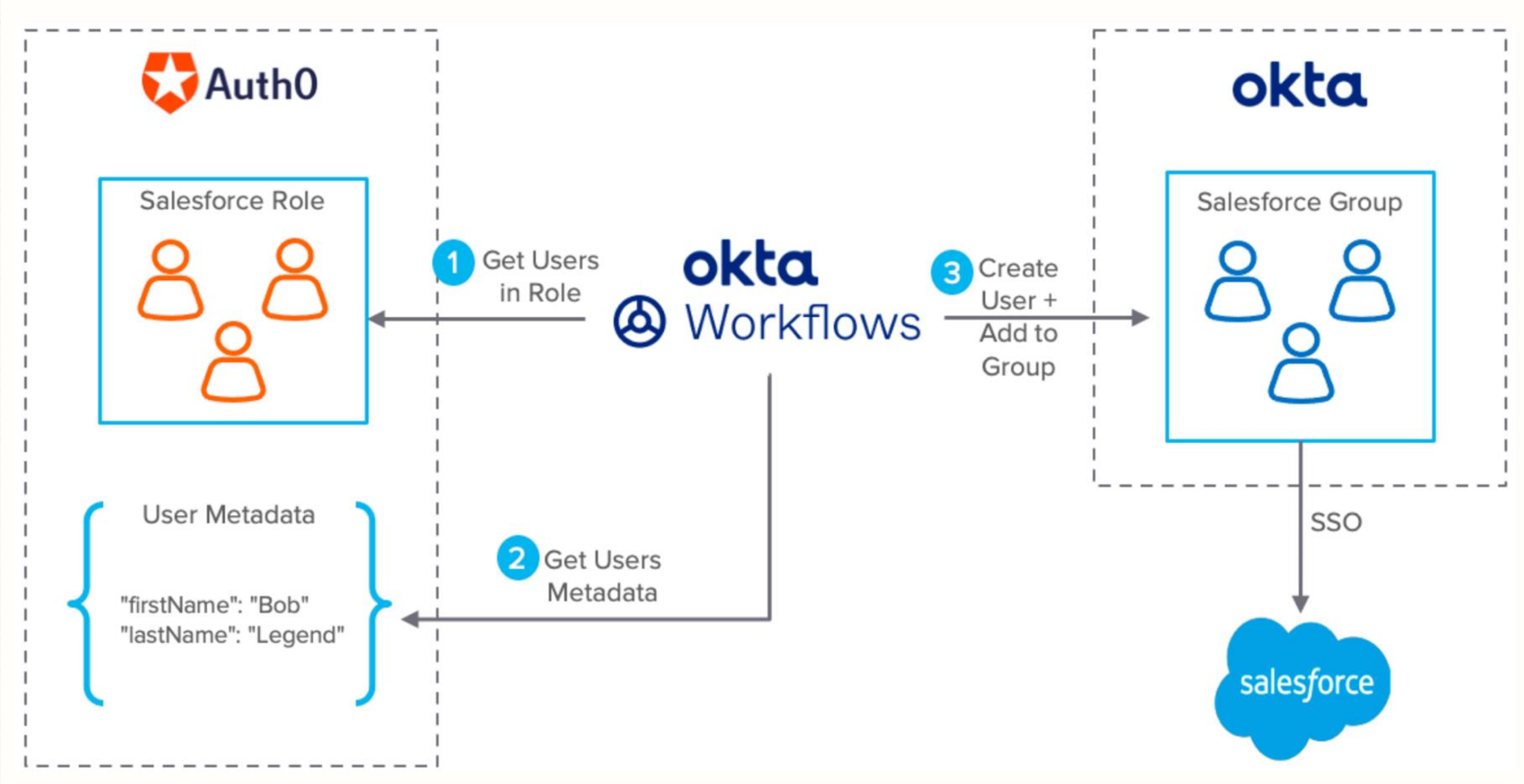
16.6 Key Terms, Review Questions, and Problems

User Authentication

- **User Authentication** is the process of determining whether some user or some application or process acting on behalf of a user is, in fact, who or what I declares itself to be.

- **Authentication technology** provides access control for systems by checking to see if a user's credentials match the credentials in a database of authorized users or in a data authentication server.

User Authentication Example



Means of Authentication

- **Authentication factors** of authenticating a user's identity:
- Knowledge factor (something you know)
  - Requires the user to demonstrate knowledge of secret information. (pass, PIN, ID number, etc.)
- **Possession Factor** (something the individual has)
  - Hardware token, badge, etc.
- **Inherence factor** (something individual is or does)
  - **Biometrics**
    - Static
    - Dynamic

Authentication Factors

- **Authenticators –** specific items used during authentication, such as password or hardware token.

Table 16.1 Authentication Factors

Factor	Examples	Properties
Knowledge	User ID Password PIN	Can be shared Many passwords easy to guess Can be forgotten
Possession	Smart Card Electronic Badge Electronic Key	Can be shared Can be duplicated (cloned) Can be lost or stolen
Inherence	Fingerprint Face Iris Voice print	Not possible to share False positives and false negatives possible Forging difficult

Multifactor Authentication

- **MFA** – use of more than one of the authentication means in the preceding list (i.e., PIN and hardware token, etc.)
- MFA is more secure than single factor authentication

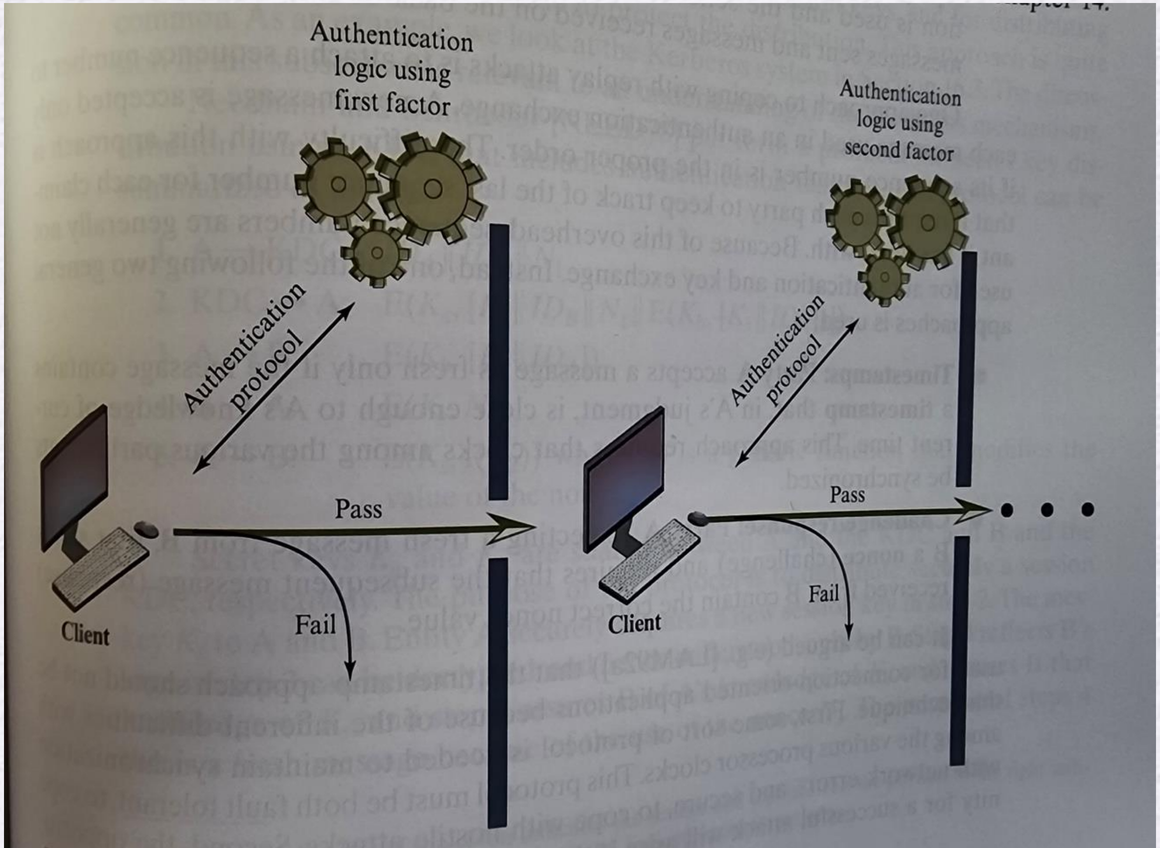
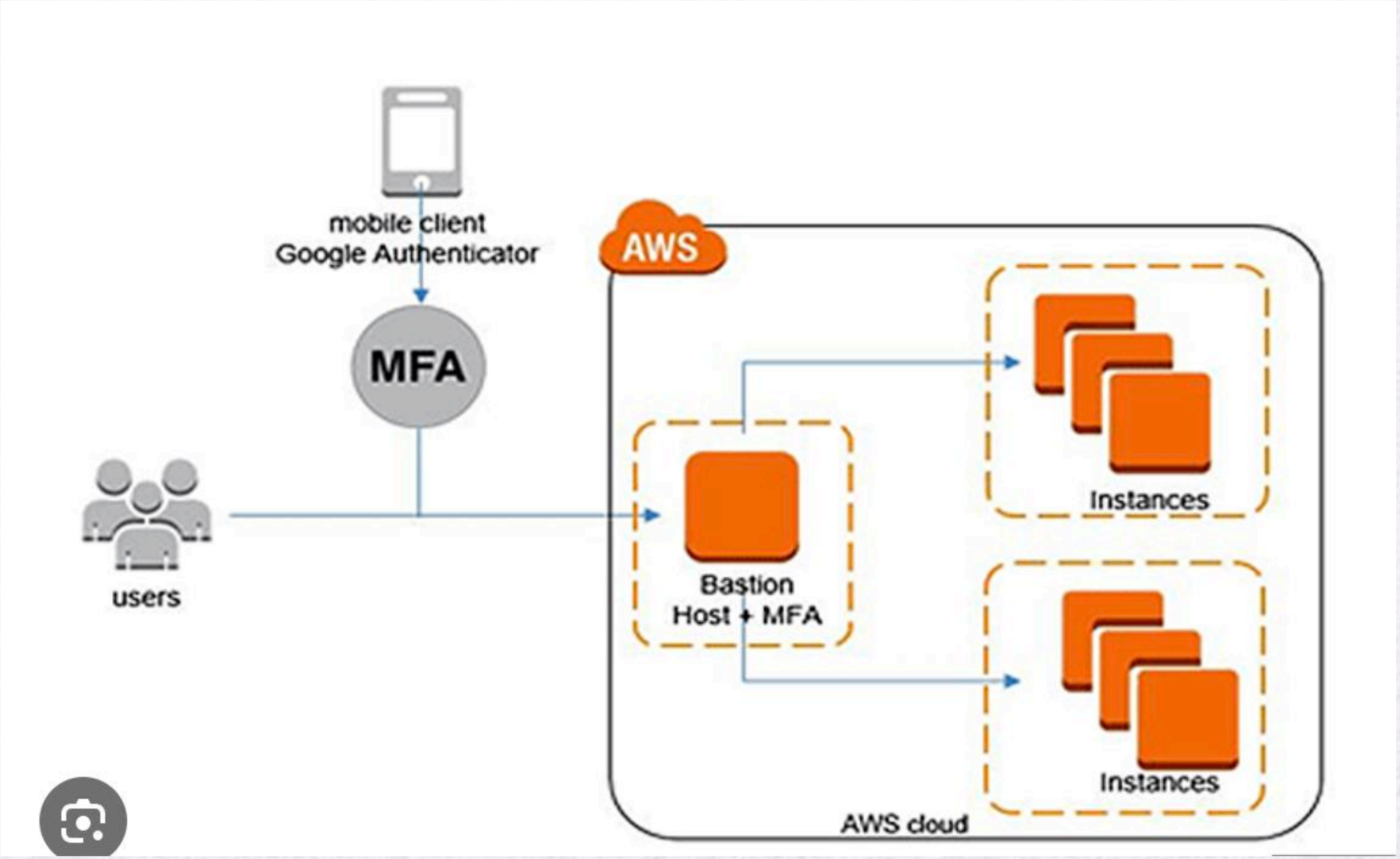


Figure 16.2 Multifactor Authentication

Multifactor Authentication - AWS



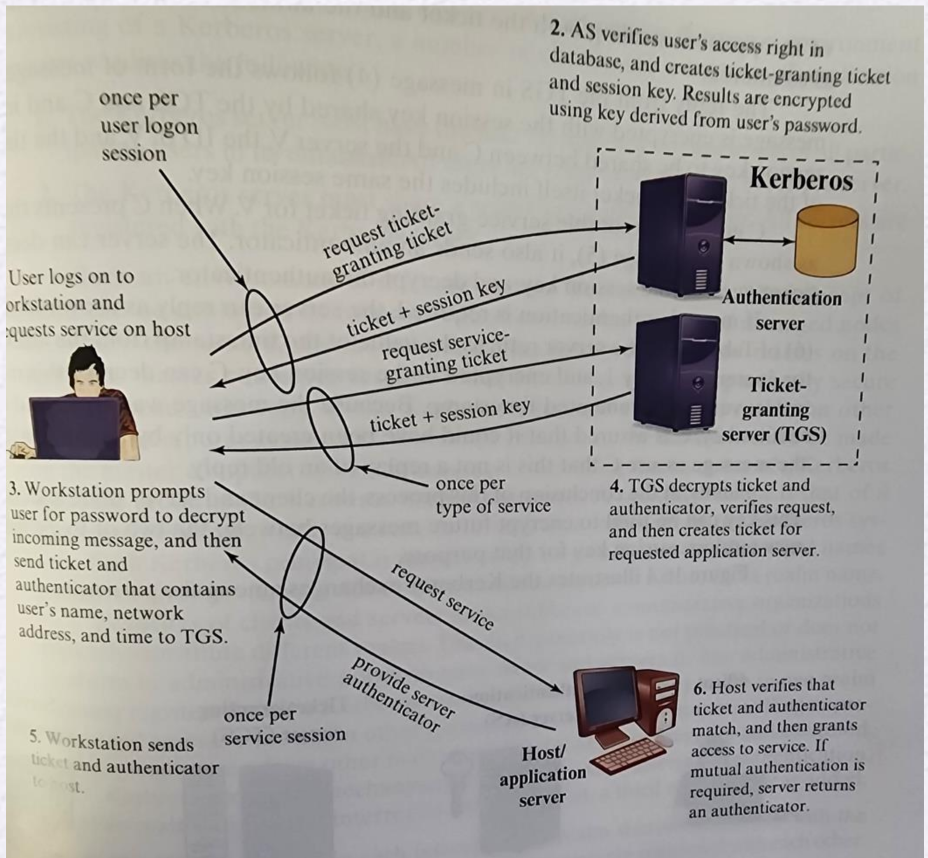
Remote User Authentication – Symmetric Keys

- Strategy involves the use of trusted key distribution center (KDC) to provide confidentiality.
- Each party in the network shares a secret key, known as master key , with the KDC.
- KDC generates keys to be used for a short time over a connection between two parties, known as session keys, and from distributing those keys using the master keys to protect the distribution. This approach is commonly used in Kerberos protocol

Kerberos Protocol

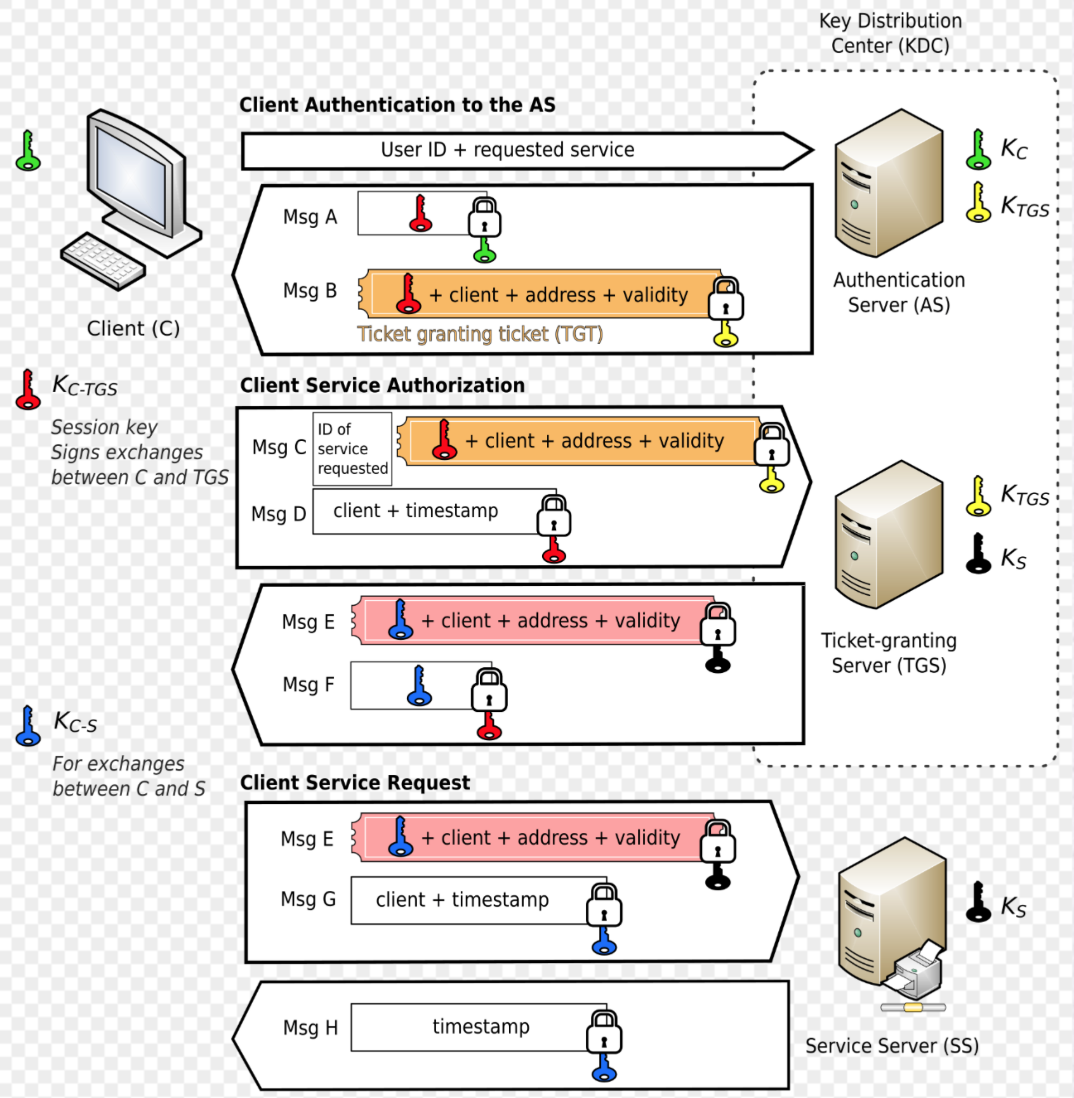
- Kerberos – authentication service that supports open distributed environment in which users at workstations wish to access services on servers distributed throughout the network.
- Kerberos provides a centralized **authentication server** whose function is to authenticate users to servers and servers to users.
- Kerberos relies almost exclusively on Symmetric encryption, and rarely on Public Key encryption.
- Current version 5

Kerberos Protocol



re 16.3 Overview of Kerberos

Kerberos Protocol with Symmetric keys



https://en.wikipedia.org/wiki/Kerberos_(protocol)#:~:text=MIT%20makes%20an%20implementation%20of,CyberSafe%20offering%20commercially%20supported%20versions.

Federated Identity Management

- Common identity management scheme across multiple enterprises and numerous applications.
- Identity Management is a centralized, automated approach to provide enterprise-wide access to resources by employees and other authorized individuals.
- The focus of identity management is defining an identity for each user, associating attributes with the identity, and enforcing a means by which a user can verify identity.
- The central concept of an identity management system is the use of single sign-on (SSO)
- SSO and MFA could be combined together (Okta, Due, etc.).

Federated Identity Management

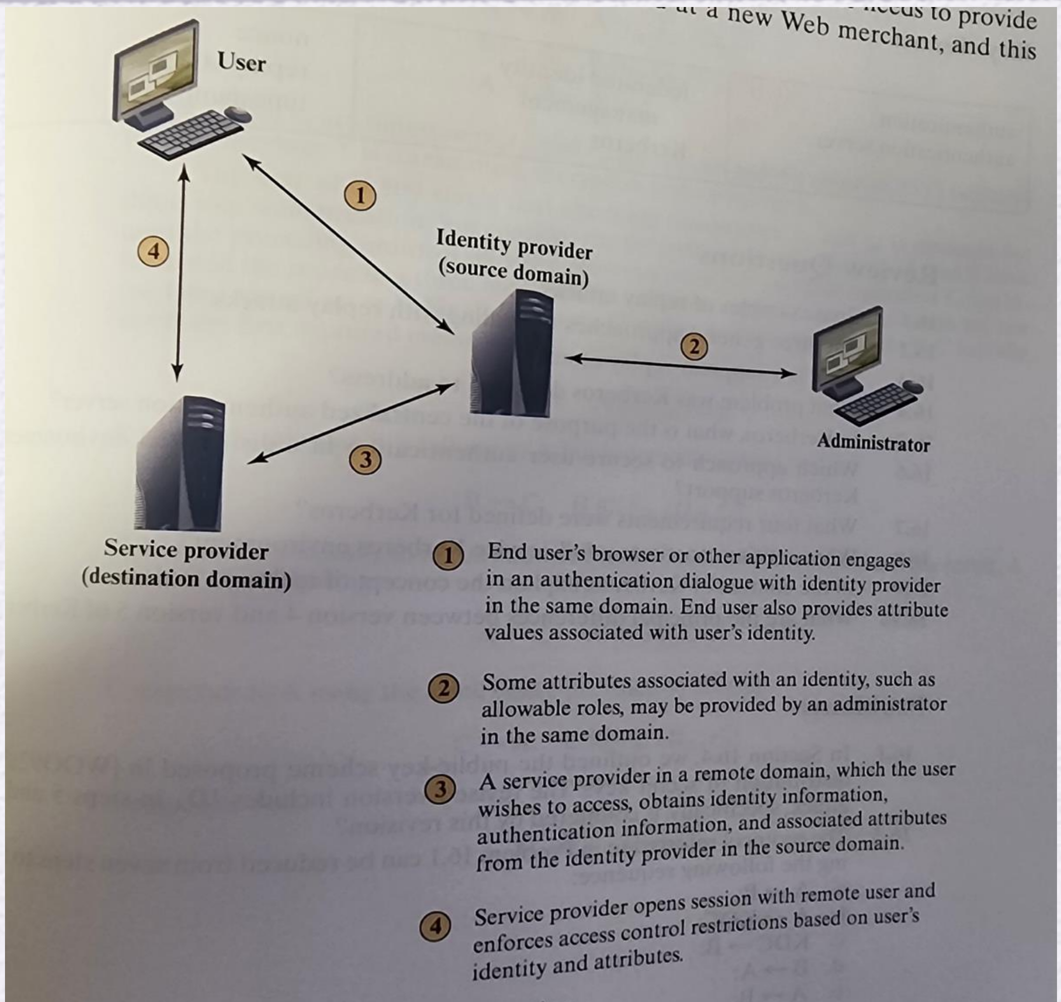


Figure 16.7 Federated Identity Operation